EXPERT SYSTEM

How computers gain expertise?

What is an EXPERT SYSTEM?

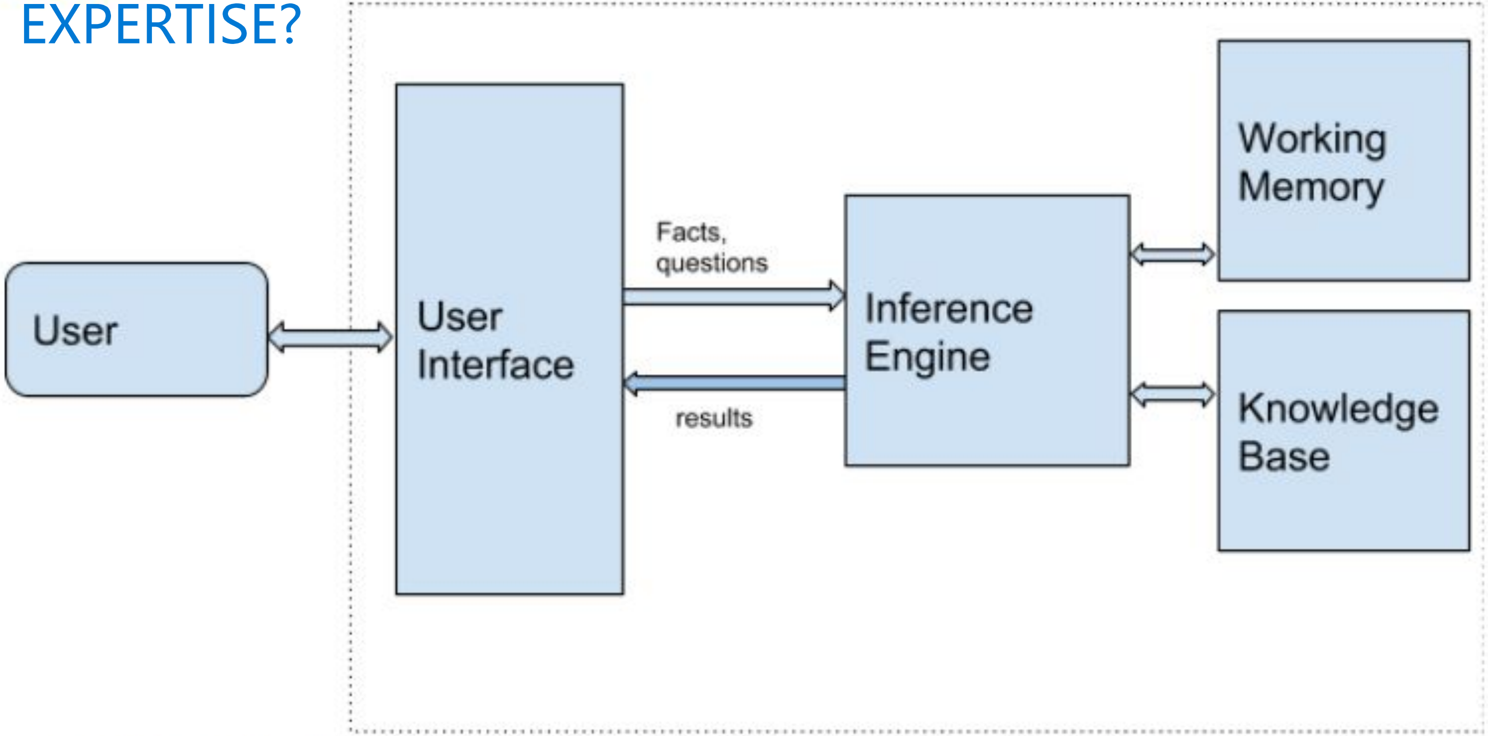
- An expert system is a computer expert that emulates the decision making ability of a human expert.
- Expert system, a computer program that uses artificial intelligence methods to solve complex problems within a specialized domain that ordinarily requires human expertise.

Example: Making of a medical diagnosis expert system

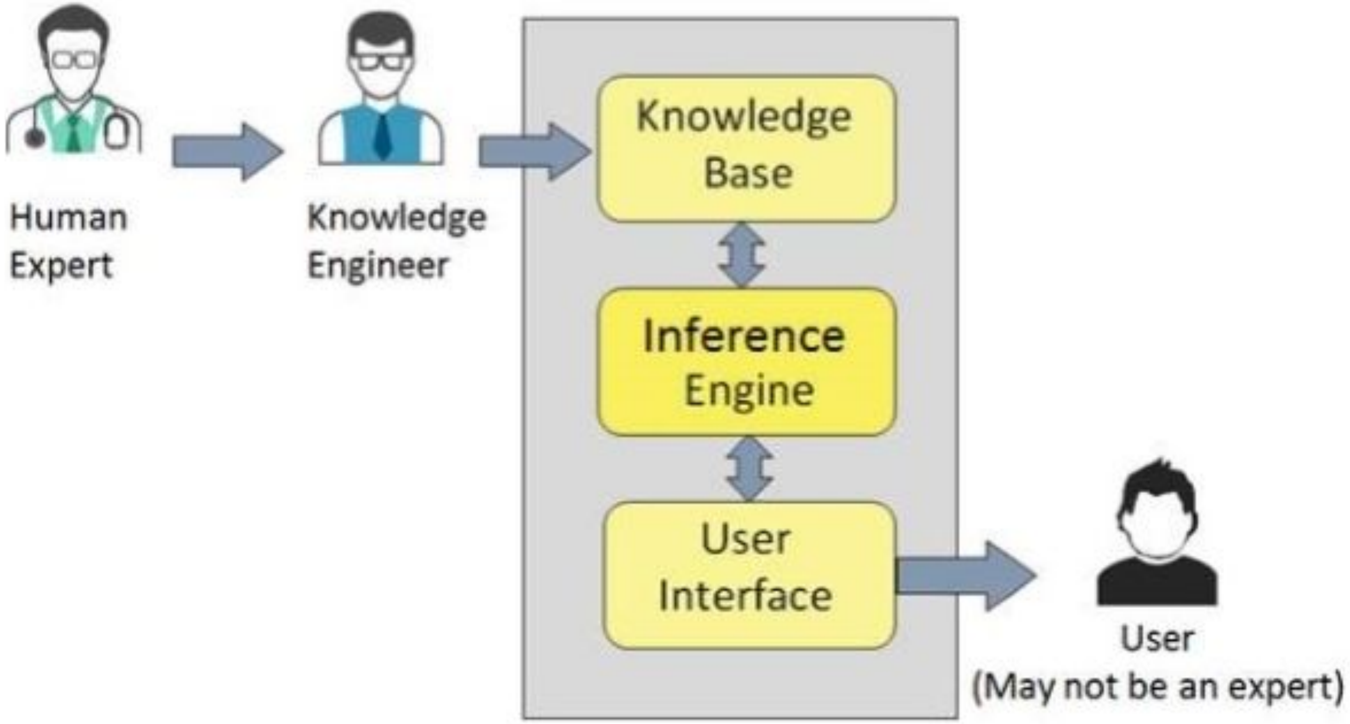
- A medical diagnosis expert system lets the user diagnose his disease without going to a real doctor (but of course, user has to go to a real doctor for treatment)

How does the EXPERT SYSTEM gain EXPERTISE?

EXPERTISE?



How does medical diagnosis expert system gain expertise?



Components of an Expert System

1. Knowledge Base
2. Inference Engine
3. User Interface

1. Knowledge Base

- **A knowledge base is an organized collection of facts about the system's domain.**
- Facts for a knowledge base must be acquired from human experts through interviews and observations.
- This knowledge is then usually represented in the form of "if-then" rules (**production rules**): "If some condition is true, then the following inference can be made (or some action taken)."
- **A probability factor** is often attached to the conclusion of each production rule and to the ultimate recommendation, because the conclusion is not a certainty.

Components of Knowledge Base

- **Factual Knowledge** – It is the information widely accepted by the Knowledge Engineers and scholars in the task domain.
- **Heuristic Knowledge** – It is about practice, accurate judgement, one's ability of evaluation, and guessing.

Knowledge Acquisition

- The knowledge base is formed by readings from various experts, scholars, and **the Knowledge Engineers**. The knowledge engineer is a person with the qualities of empathy, quick learning, and **case analyzing skills.**

- He acquires information from **subject expert** by recording, interviewing, and observing him at work, etc.

- He then categorizes and organizes the information in a meaningful way, in the form of **IF-THEN-ELSE rules**, to be used by interference machine. The knowledge engineer also monitors the development of the ES.

Knowledge Base for Medical Diagnosis Expert system

Symptoms	Disease
Fever, cough, conjunctivitis, running nose, rashes	Measles
Fever,headache, ache, conjunctivitis, chills,sore throat,cough,running nose	Flu
Headache, sneezing, sore throat, running nose, chills	Cold
Fever, chills,ache,rashes	Chickenpox

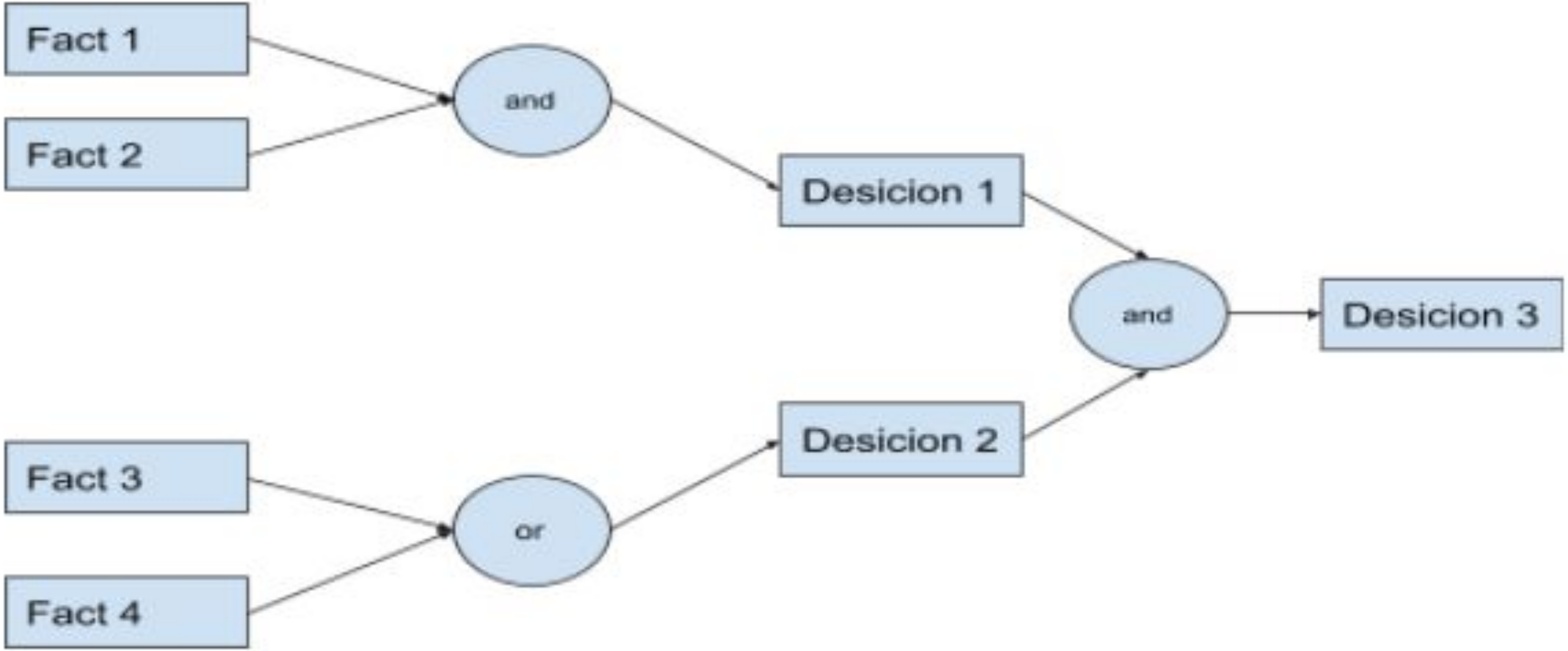
2. The Inference Engine

- **An inference engine interprets and evaluates the facts in the knowledge base in order to provide an answer or to reach a goal state**
- For example, if the KB contains the production rules “if *x*, then *y*” and “if *y*, then *z*,” the inference engine is able to deduce “if *x*, then *z*.”
- In medical diagnosis ES, i**f a user has Fever, chills,ache,rashes, then user is suffering from chickenpox**
- Strategies used by the inference engine
  - Forward chaining
  - Backward chaining

Forward Chaining

- **“What can happen next?”**
- It is a data-driven strategy.
- The inferencing process moves from the facts of the case to a goal (conclusion).
- The inference engine attempts to match the condition (IF) part of each rule in the knowledge base with the facts currently available in the working memory.
- If several rules match, a conflict resolution procedure is invoked;

Forward chaining diagram



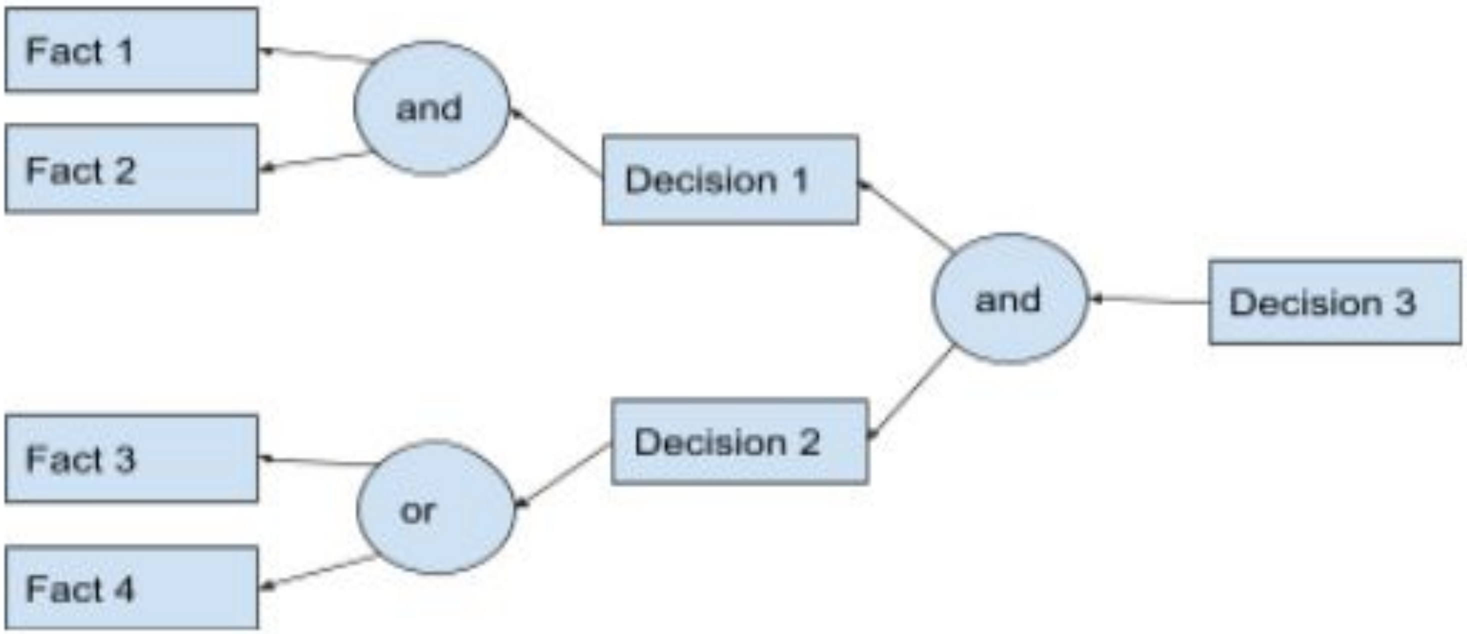
Forward Chaining in Medical Diagnosis ES

- A user has fever, headache, ache, conjunctivitis, chills,sore throat,cough,running nose.
- In forward chaining, the inference engine will use the symptoms specified to conclude to the disease the user is suffering from.
- After going through the above symptoms, the inference engine concludes that the user is suffering from flu by checking the knowledge base.

Backward Chaining

- **“Why this happened?”**
- On the basis of what has already happened, **the Inference Engine tries to find out which conditions could have happened in the past for this result.**
- This strategy is followed for finding out cause or reason.
- The inference engine attempts to match the assumed (hypothesized) conclusion - the goal or subgoal state - with the conclusion (THEN) part of the rule.
- If such a rule is found, its premise becomes the new subgoal.
- In an ES with few possible goal states, this is a good strategy to pursue.

Backward Chaining Diagram



Backward Chaining in Medical Diagnosis ES

- Suppose a user has measles.
- Now, the inference engine will look up in the knowledge base to derive the symptoms that could have resulted in measles.
- It inferred that the symptoms were fever, cough, conjunctivitis, running nose, rashes.
- This worked as only few states were present.

3. User Interface

- User interface provides interaction between user of the ES and the ES itself.
- The user of the ES need not be necessarily an expert in Artificial Intelligence.
- It explains how the ES has arrived at a particular recommendation. The explanation may appear in the following forms –
  - Natural language displayed on screen.
  - Verbal narrations in natural language.
  - Listing of rule numbers displayed on the screen.